BJT Amplifiers

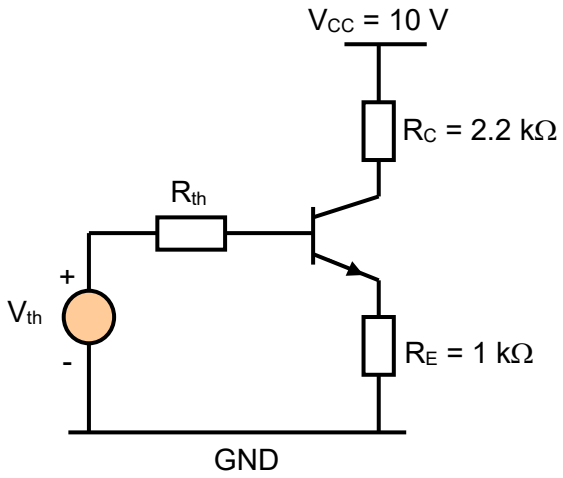
Tutorial Solutions

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Question 1: Common-Emitter Amplifier with Negative Feedback

We use Thévenin's theorem to simplify the DC analysis. For the DC mode, the amplifier circuit is thus equivalent to the circuit depicted below, with

$$V_{th} = \frac{R_2 V_{CC}}{R_1 + R_2} \approx 2.7 \text{ volts and } R_{th} = \frac{R_1 R_2}{R_1 + R_2} \approx 7.3 \text{ k}\Omega.$$



We assume that the BJT is in the forward active mode (to be checked later). Under this assumption, we show that

$$V_{th} \approx R_{th} I_{B0} + V_{BE,on} + R_E I_{E0} = R_{th} I_{B0} + V_{BE,on} + R_E (1+\beta_F) I_{B0} \approx [R_{th} + R_E \beta_F] I_{B0} + V_{BE,on}$$

$$\Rightarrow \quad I_{B0} \approx \frac{V_{th} - V_{BE,on}}{R_{th} + R_E \beta_F} \approx 18.6 \ \mu\text{A}.$$

$$\Rightarrow \quad I_{C0} = \beta_F I_{B0} \approx 1.86 \text{ mA and } I_{E0} = (1+\beta_F) I_{B0} \approx I_{C0} \approx 1.86 \text{ mA}.$$

Note that the DC collector current can be expressed as $I_{C0} \approx \beta_F \frac{V_{th} - V_{BE,on}}{R_{th} + R_E \beta_F}$.

This equation is particularly interesting because it indicates that the value of β_F has very little effect on the DC operating point.

The excellent stability of the DC operating point is actually due to the use of negative feedback. In fact, without feedback ($R_E = 0$), we would obtain

$$I_{C0} \approx \beta_F \frac{V_{th} - V_{BE,on}}{R_{th}},$$

thus implying that the DC operating point would be very much dependent on the value of β_F.

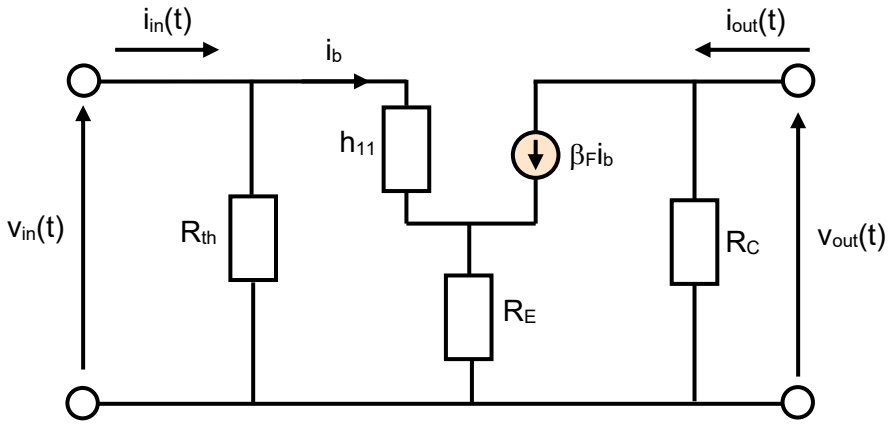
We also have $V_{CC} = R_C I_{C0} + V_{CE0} + R_E I_{E0} \approx (R_C + R_E) I_{C0} + V_{CE0}$, which yields

$$V_{CE0} \approx V_{CC} - (R_C + R_E) I_{C0} \approx 4.1 \text{ volts}.$$

The transistor is thus in the forward active mode.

For a CE amplifier with a single voltage supply V_{CC}, the value of V_{CE0} that maximises the output voltage swing is approximately equal to $V_{CC}/2 = 5$ volts. Since we found $V_{CE0} \approx 4.1$ volts here, we conclude that the BJT is properly (but not optimally) biased.

As for the AC analysis, the small-signal model of our feedback amplifier is depicted below.



The general equations for this circuit are as follows:

(1) $v_{in}(t) = R_{th}(i_{in}(t) - i_b)$, (2) $v_{in}(t) = [h_{11} + (\beta_F + 1)R_E]i_b \approx (h_{11} + \beta_F R_E) i_b$,

and (3) $v_{out}(t) = R_C(i_{out}(t) - \beta_F i_b)$.

• OC Voltage Gain: The OC voltage gain of our feedback CE amplifier is given by

$$A_v = \left.\frac{v_{out}(t)}{v_{in}(t)}\right|_{i_{out}(t)=0} \approx -\frac{\beta_F R_C}{h_{11} + R_E \beta_F}.$$

At room temperature, we can write $h_{11} \approx \frac{25\text{ mV}}{I_{B0}}$, with $I_{B0} \approx 18.6\ \mu A$. So, we have $h_{11} \approx 1.34$ kΩ. Since $h_{11} << R_E \beta_F$, we finally have $A_v \approx -\frac{R_C}{R_E} \approx$ -2.2.

Without feedback ($R_E = 0$), we would obtain

$$A_v \approx -\beta_F \frac{R_C}{h_{11}},$$

thus resulting in a much higher voltage gain equal to -163. Negative feedback does significantly reduce the gain of an amplifier, but the good thing is that this gain does not depend on the value of β_F and is therefore very stable.

• Input Resistance: The input resistance is given by

$$R_{in} = \frac{v_{in}(t)}{i_{in}(t)} = \frac{[h_{11} + R_E (1+\beta_F)] R_{th}}{h_{11} + R_E (1+\beta_F) + R_{th}} \approx \frac{R_E \beta_F R_{th}}{R_E \beta_F + R_{th}} \approx 6.8\ \text{k}\Omega.$$

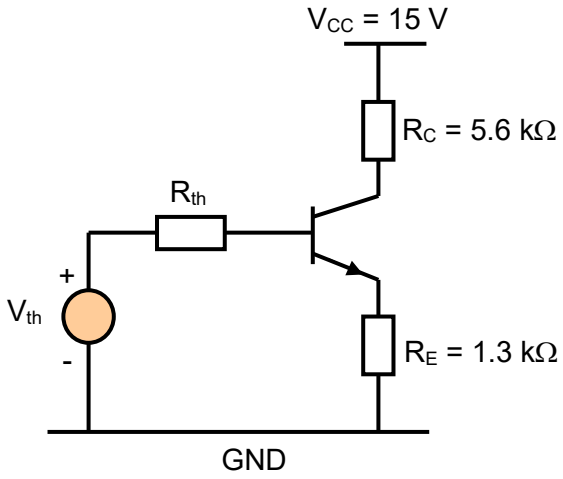
Without feedback ($R_E = 0$), we would obtain $R_{in} = \frac{h_{11} R_{th}}{h_{11} + R_{th}} \approx 1.1$ kΩ. So, negative feedback has increased the input resistance, which is really good!

• Output Resistance: The output resistance is given by $R_{out} = \left.\frac{v_{out}(t)}{i_{out}(t)}\right|_{v_{in}(t)=0} = R_C$ = 2.2 kΩ (same result as without feedback).

Question 2: CE Amplifier with Negative Feedback and Decoupling Capacitor

We use Thévenin's theorem to simplify the DC analysis. For the DC mode, the amplifier circuit is thus equivalent to the circuit depicted below, with

$$V_{th} = \frac{R_2 V_{CC}}{R_1 + R_2} \approx 2.05 \text{ volts and } R_{th} = \frac{R_1 R_2}{R_1 + R_2} \approx 11.2 \text{ k}\Omega.$$



Now, we assume that the BJT is in the forward active mode (to be checked later). Under this assumption, we show that

$$V_{th} = R_{th} I_{B0} + V_{BE,on} + R_E I_{E0} = R_{th} I_{B0} + V_{BE,on} + R_E (1+\beta_F) I_{B0} \approx (R_{th} + R_E \beta_F) I_{B0} + V_{BE,on}$$

$$\Rightarrow I_{B0} \approx \frac{V_{th} - V_{BE,on}}{R_{th} + R_E \beta_F} \approx 9.6 \ \mu\text{A}.$$

$$\Rightarrow I_{C0} = \beta_F I_{B0} \approx 0.96 \text{ mA and } I_{E0} = (1+\beta_F) I_{B0} \approx I_{C0} = 0.96 \text{ mA}.$$

Note that $I_{C0} \approx \beta_F \frac{V_{th} - V_{BE,on}}{R_{th} + R_E \beta_F}$.

This equation indicates that the value of β_F has very little effect on the DC operating point.

The excellent stability of the DC operating point is due to the use of negative feedback. In fact, without feedback (R_E = 0), we would obtain

$$I_{C0} \approx \beta_F \frac{V_{th} - V_{BE,on}}{R_{th}},$$

thus implying that the DC operating point would be very much dependent on the value of β_F.

We also have $V_{CC} = R_C I_{C0} + V_{CE0} + R_E I_{E0} \approx (R_C + R_E) I_{C0} + V_{CE0}$, which is equivalent to

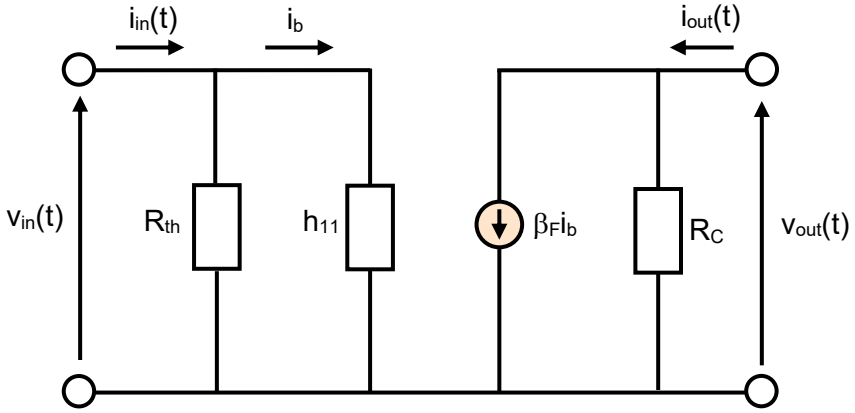
$$V_{CE0} \approx V_{CC} - (R_C + R_E) I_{C0} \approx 8.4 \text{ volts.}$$

The transistor is thus in the forward active mode.

For a CE amplifier with a single voltage supply V_{CC}, the value of V_{CE0} that maximises the output voltage swing is approximately equal to $V_{CC}/2 = 7.5$ volts. Since we found $V_{CE0} \approx 8.4$ volts here, we conclude that the BJT is properly (but not optimally) biased.

As for the AC analysis, the small-signal model of our amplifier is depicted below. We have assumed that the decoupling capacitance is equivalent to a short circuit at the frequency of the input signal.

We thus obtain the small-signal model of a classical CE amplifier similar to that described in the lecture notes.



The general equations for this circuit are as follows:
(1) $v_{in}(t) = R_{th}\,(i_{in}(t) - i_b)$, (2) $v_{in}(t) = h_{11} i_b$, and (3) $v_{out}(t) = R_C\,(i_{out}(t) - \beta_F\, i_b)$.

• OC Voltage Gain: The OC voltage gain of the CE amplifier is given by

$$A_v = \left.\frac{v_{out}(t)}{v_{in}(t)}\right|_{i_{out}(t)=0} = -\beta_F \frac{R_C}{h_{11}}.$$

At room temperature, we have $h_{11} \approx \dfrac{25 \text{ mV}}{I_{B0}}$, with $I_{B0} \approx 9.6\ \mu\text{A}$. So, we have $h_{11} \approx 2.6\ \text{k}\Omega$.

We can finally write $A_v = -\beta_F \frac{R_C}{h_{11}} \approx$ -215,

which is a much higher voltage gain than that achieved in Question 1, thanks to the decoupling capacitance. Note that this gain depends on the value of β_F and is therefore unstable.

- Input Resistance: The input resistance is given by

$$R_{in} = \frac{v_{in}(t)}{i_{in}(t)} = \frac{h_{11} R_{th}}{h_{11} + R_{th}} \approx 2.1\ \text{k}\Omega.$$

- Output Resistance: The output resistance is given by

$$R_{out} = \left.\frac{v_{out}(t)}{i_{out}(t)}\right|_{v_{in}(t)=0} = R_C = 5.6\ \text{k}\Omega.$$

Question 3: Cascaded Amplifier

The real voltage gain of the cascaded amplifier is given by

$$\tilde{A}_v = (A_v)^2 \frac{R_{in}}{R_{in} + R_s} \frac{R_{in}}{R_{in} + R_{out}} \frac{R_L}{R_L + R_{out}},$$

with $A_v = -\beta_F \frac{R_C}{h_{11}} \approx$ -215, $R_{in} = \frac{h_{11} R_{th}}{h_{11} + R_{th}} \approx 2.1\ \text{k}\Omega$, and $R_{out} = R_C = 5.6\ \text{k}\Omega$.

Hence, we have

$$\tilde{A}_v \approx (-215)^2 \times \frac{2.1}{2.1+1} \times \frac{2.1}{2.1+5.6} \times \frac{10}{10+5.6} \approx 46{,}225 \times 0.677 \times 0.273 \times 0.641 \approx 5476.$$

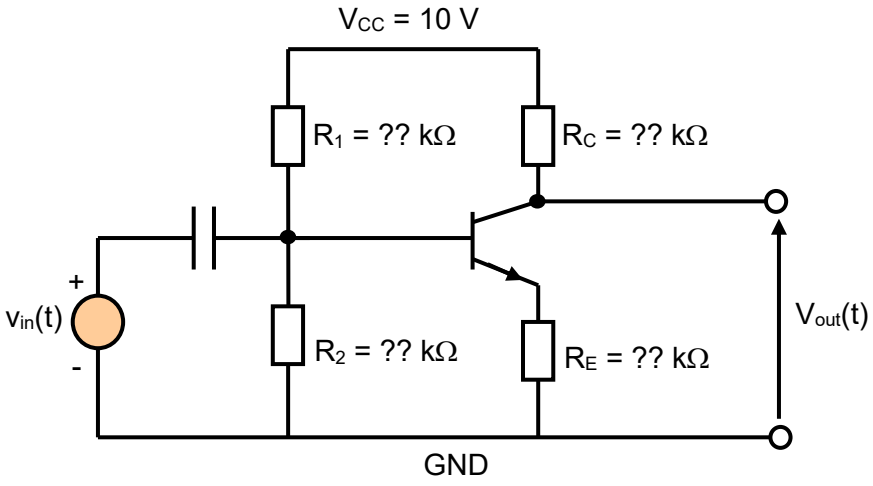
If the load resistance R_L was equal to 10 Ω instead of 10 kΩ, the voltage gain would be

$$\tilde{A}_v \approx (-215)^2 \times \frac{2.1}{2.1+1} \times \frac{2.1}{2.1+5.6} \times \frac{0.01}{0.01+5.6} \approx 46{,}225 \times 0.677 \times 0.273 \times 0.00178 \approx 15.2.$$

The real voltage gain of the cascaded amplifier would thus be significantly reduced. We conclude that the real voltage gain strongly depends on the input resistance R_L of the loading circuit. This is due to the rather high output resistance R_{out} = 5.6 kΩ of this amplifier.

Question 4: Design of a CE Amplifier with Negative Feedback

The structure of our amplifier is depicted below.



The design issue consists of determining the values of the four resistances R_1, R_2, R_C and R_E so as to meet the specifications. To this end, we must remember several results that have previously been obtained for the CE amplifier with negative feedback.

We start with the expression of the OC voltage gain:

$$A_v = \left.\frac{v_{out}(t)}{v_{in}(t)}\right|_{i_{out}(t)=0} \approx -\frac{\beta_F R_C}{h_{11} + R_E \beta_F},$$

which can simplified as $A_v \approx -\dfrac{R_C}{R_E}$ as long as $h_{11} << R_E \beta_F$.

Assume that we indeed have $h_{11} << R_E \beta_F$ (to be checked later). In this case, a voltage gain of approximately -9 can be achieved by choosing, for example, R_C = 10 kΩ and R_E = 1 kΩ (thus leading to $A_v \approx$ -10, which gives us a bit of allowance).

Note that the output resistance R_{out} of the amplifier is given by $R_{out} = R_C$ = 10 kΩ. This value tends to be a bit high for an output resistance but can still considered as acceptable.

Obviously, we could reduce both R_C and R_E in order to decrease the output resistance while keeping the voltage gain unchanged. However, doing so would negatively impact some other parameters in this circuit. For instance, if R_E becomes too low, the condition $h_{11} << R_E \beta_F$ may not hold anymore. Ultimately, there are several conflicting constraints to consider when designing such a circuit, and a trade-off must be found.

Now, we are also required to maximise the output voltage swing. We have seen in class that, for a CE amplifier with a single voltage supply V_{CC}, the value of V_{CE0} that maximises the output voltage swing is approximately equal to $V_{CC}/2 = 5$ volts.

Using the following equations:
$V_{CC} = R_C I_c(t) + R_E I_e(t) + V_{ce}(t) \approx (R_C + R_E) I_c(t) + V_{ce}(t)$ and $V_{CC} = R_C I_c(t) + V_{out}(t)$,
we show that the optimal value $V_{CE0} \approx V_{CC}/2$ corresponds to

(1) $I_{C0} \approx \dfrac{V_{CC}}{2(R_C + R_E)} \approx 0.45$ mA,

(2) $V_{OUT,0} \approx \dfrac{V_{CC}}{2}\left[\dfrac{R_C + 2R_E}{R_C + R_E}\right] \approx 5.45$ V.

Here, we are required to specify the value of the output voltage swing. We can proceed as explained below.

We must first find “where the cut-off mode starts” and “where the saturation mode starts”.

Let us use the following notations for the compound signals (AC + DC) in the circuit: $I_c(t) = i_c(t) + I_{C0}$, $V_{ce}(t) = v_{ce}(t) + V_{CE0}$ and $V_{out}(t) = v_{out}(t) + V_{OUT,0}$.

The cut-off mode of operation corresponds to a collector current being equal to zero: $I_c(t) = 0$. Using the following general equations:
$V_{CC} = R_C I_c(t) + R_E I_e(t) + V_{ce}(t) \approx (R_C + R_E) I_c(t) + V_{ce}(t)$ and $V_{CC} = R_C I_c(t) + V_{out}(t)$, we can see that the BJT enters the cut-off mode once we have: (1) $I_c(t) = 0$, (2) $V_{ce}(t) = V_{CC}$, and (3) $V_{out}(t) = V_{CC}$.

The saturation mode of operation corresponds to a voltage $V_{ce}(t)$ being equal to $V_{CE,sat} \approx$ 0.2 volt. Using the following general equations:

$V_{CC} = R_C I_c(t) + R_E I_e(t) + V_{ce}(t) \approx (R_C + R_E) I_c(t) + V_{ce}(t)$ and $V_{CC} = R_C I_c(t) + V_{out}(t)$, we can see that the BJT enters the saturation mode once we have

(1) $I_c(t) \approx \dfrac{V_{CC} - V_{CE,sat}}{R_C + R_E} \approx \dfrac{V_{CC}}{R_C + R_E}$, (2) $V_{ce}(t) = V_{CE,sat}$,

and (3) $V_{out}(t) \approx V_{CC} - \dfrac{R_C V_{CC}}{R_C + R_E} = \dfrac{R_E V_{CC}}{R_C + R_E}$.

The value of the output voltage swing is given by the difference between the value of $V_{out}(t)$ corresponding to the start of the cut-off region and that corresponding to the start of the saturation region. Thus, this voltage swing is given by

$$V_{CC} - \frac{R_E V_{CC}}{R_C + R_E} = \frac{R_C V_{CC}}{R_C + R_E} \approx 9.1 \text{ volts.}$$

Now, let us proceed with the remaining of the DC analysis.

The value of the DC collector current I_{C0} leads to the desired value for the base current:

$$I_{B0} = \frac{I_{C0}}{\beta_F} \approx 4.5 \ \mu\text{A.}$$

Note that we can now compute the resistance h_{11} at room temperature as follows:

$$h_{11} \approx \frac{V_T}{I_{B0}} \approx \frac{25 \text{ mV}}{4.5 \ \mu\text{A}} \approx 5.6 \ \text{k}\Omega.$$

This confirms that the assumption $h_{11} << R_E \beta_F$ previously made can be considered as valid (as 5.6 kΩ << 100 kΩ).

We remember that the expression of the DC base current is given by

$$I_{B0} \approx \frac{V_{th} - V_{BE,on}}{R_{th} + R_E \beta_F} \approx 4.5 \ \mu\text{A.}$$

Hence, we must choose the values of the quantities $V_{th} = \frac{R_2 V_{CC}}{R_1 + R_2}$ and $R_{th} = \frac{R_1 R_2}{R_1 + R_2}$ so that $I_{B0} \approx 4.5\ \mu A$. There are obviously several possible choices.

We also remember that the input resistance of the CE amplifier with negative feedback is given by

$$R_{in} = \frac{[h_{11} + R_E (1+\beta_F)] R_{th}}{h_{11} + R_E (1+\beta_F) + R_{th}} \approx \frac{R_E \beta_F R_{th}}{R_E \beta_F + R_{th}}.$$

This expression clearly shows that, in order to obtain an input resistance R_{in} that is sufficiently large, we must make sure to select both resistances R_1 and R_2 so that R_{th} is not too small.

On the other hand, R_{th} should not be too large either because it is crucial that the DC operating point does not significantly depend on the parameter β_F; indeed remember that the collector current is given by

$$I_{C0} \approx \beta_F \frac{V_{th} - V_{BE,on}}{R_{th} + R_E \beta_F}.$$

If $R_{th} << R_E \beta_F$, we have

$$I_{C0} \approx \frac{V_{th} - V_{BE,on}}{R_E},$$

which is highly desirable in practice since the collector current is then completely independent from the parameter β_F.

We clearly have conflicting constraints once again!

In general, the stability of the DC operating point is the most crucial issue, i.e. we ensure that the condition $R_{th} << R_E \beta_F$ is satisfied.

In this case, we can write $V_{th} \approx V_{BE,on} + R_E I_{C0} \approx 0.7\ V + 1 k\Omega \times 0.45\ mA \approx 1.15$ volt. In other words, we must select both resistances R_1 and R_2 so that

$$V_{th} = \frac{R_2 V_{CC}}{R_1 + R_2} = \frac{R_2}{R_1 + R_2} \times 10 \text{ V} \approx 1.15 \text{ volt},$$

under the constraint $R_{th} = \frac{R_1 R_2}{R_1 + R_2} << R_E \beta_F = 100 \text{ k}\Omega$.

We thus have $\frac{R_2}{R_1 + R_2} \approx 0.115$, which yields $R_2 \approx 0.13 \times R_1$, with R_2 being equal, typically, to a few kΩs. For instance, we can choose R_2 = 3 kΩ and R_1 = 23 kΩ.

Final checks (does it work?):

We have chosen the following values for the four resistances: R_1 = 23 kΩ, R_2 = 3 kΩ, R_C = 10 kΩ, and R_E = 1 kΩ.

With these values, we obtain

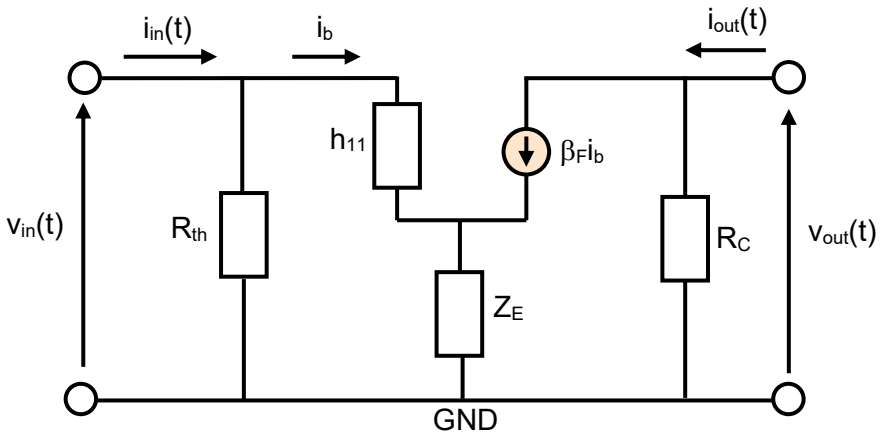
- $V_{th} = \frac{R_2 V_{CC}}{R_1 + R_2} \approx 1.15$ volts;
- $R_{th} = \frac{R_1 R_2}{R_1 + R_2} \approx 2.65$ kΩ.

This result confirms that $R_{th} << R_E \beta_F = 100 \text{ k}\Omega$, thus guaranteeing an excellent stability of the DC operating point;

- $I_{C0} \approx \beta_F \frac{V_{th} - V_{BE,on}}{R_{th} + R_E \beta_F} \approx 0.44$ mA (very close to the desired 0.45 mA) and, thus, $I_{B0} = \frac{I_{C0}}{\beta_F} \approx$ 4.4 μA (very close to the desired 4.5 μA);
- $V_{CE0} \approx V_{CC} - (R_C + R_E) I_{C0} \approx 5.2$ volts (very close to the desired 5 volts) and $V_{OUT,0} \approx V_{CC} - R_C I_{C0} \approx 5.6$ volts (very close to the desired 5.45 volts);
- $h_{11} \approx \frac{V_T}{I_{B0}} \approx \frac{25 \text{ mV}}{4.4 \text{ μA}} \approx 5.7$ kΩ;
- $A_v \approx -\frac{\beta_F R_C}{h_{11} + R_E \beta_F} \approx$ -9.5 (very close to the desired OC voltage gain of 9);
- $R_{in} \approx \frac{[h_{11} + R_E \beta_F] R_{th}}{h_{11} + R_E \beta_F + R_{th}} \approx 2.6$ kΩ, which means that the input resistance is “not too low”;
- $R_{out} = R_C$ = 10 kΩ, which means that the output resistance is “not too large”.

Question 5: CE Amplifier with Negative Feedback and Decoupling Capacitor

The small-signal model of the amplifier circuit studied in Question 2 (CE amplifier with negative feedback and decoupling capacitor) is depicted below.



We assume that the input voltage $v_{in}(t)$ is a sinusoidal signal with frequency ω. The impedance $Z_E(\omega)$ corresponds to the capacitance C_E in parallel with the resistance R_E, i.e.

$$Z_E(\omega) = \frac{R_E \dfrac{1}{jC_E\omega}}{R_E + \dfrac{1}{jC_E\omega}} = \frac{R_E}{1 + jR_EC_E\omega}.$$

It can be shown that the OC voltage gain of this circuit is given by

$$A_v(\omega) = \left.\frac{\overline{V}_{out}}{\overline{V}_{in}}\right|_{\overline{I}_{out}=0} \approx -\frac{\beta_F R_C}{h_{11} + Z_E(\omega)\beta_F}.$$

If the frequency of the input signal is sufficient low so that we can write $\omega << \dfrac{1}{R_EC_E}$, then it appears that $Z_E(\omega) \approx R_E$ and the gain can thus be approximated as

$$A_v(\omega) \approx -\frac{\beta_F R_C}{h_{11} + R_E\beta_F}.$$

This is the same expression as for a classical CE amplifier with negative feedback. This result is not surprising because, at sufficiently low frequencies, the capacitance C_E has a very large impedance and is thus short-circuited by the resistance R_E.

As a conclusion, the use of a decoupling capacitance does not increase the OC voltage gain of a CE amplifier with negative feedback at low frequencies.

If the frequency of the input signal is sufficient high so that we can write $\omega >> \frac{1}{R_E C_E}$, then it appears that

$$Z_E(\omega) \approx \frac{R_E}{jR_E C_E \omega} = \frac{1}{jC_E \omega},$$

and the OC voltage gain can now be approximated as

$$A_v(\omega) \approx -\frac{\beta_F R_C}{h_{11} + \frac{\beta_F}{jC_E \omega}}.$$

In this case, it is clear that the resistance R_E is completely short-circuited by the decoupling capacitance C_E. As the frequency ω of the input signal is increased, the OC voltage gain progressively increases as well.

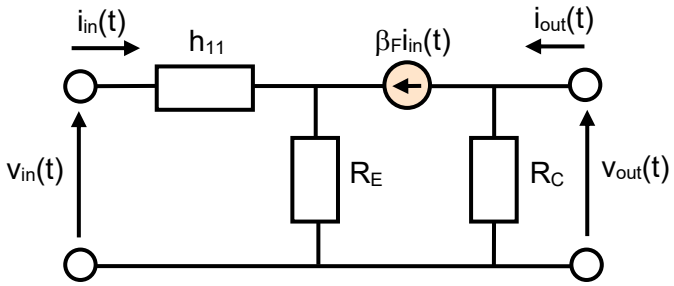
The previous expression indicates that, as $\omega \to +\infty$, we have $A_v(\omega) \to -\frac{\beta_F R_C}{h_{11}}$.

This is the expression of the gain obtained with a CE amplifier without negative feedback.

Question 6: CE Amplifier with Symmetrical Power Supplies

We are not required to perform the DC analysis of the circuit. We can simply assume that the BJT is in the forward active mode without having to demonstrate it.

As for the AC analysis, the small-signal model of our CE amplifier with symmetrical power supplies is depicted below.



The general equations for this circuit are as follows:

(1) $v_{in}(t) = [h_{11} + (\beta_F + 1)R_E]i_{in}(t) \approx (h_{11} + \beta_F R_E)i_{in}(t)$ and (2) $v_{out}(t) = R_C\,(i_{out}(t) - \beta_F\, i_{in}(t))$.

• OC Voltage Gain: The OC voltage gain of this amplifier is given by

$$A_v = \left.\frac{v_{out}(t)}{v_{in}(t)}\right|_{i_{out}(t)=0} \approx -\frac{\beta_F\, R_C}{h_{11} + R_E\,\beta_F},$$

which is the usual result obtained for a CE with negative feedback.

If we use the fact that $h_{11} << R_E\,\beta_F$, i.e. 2 kΩ << 150 kΩ, we have $A_v \approx -\dfrac{R_C}{R_E} \approx$ -4.1.

These results show that using two symmetrical power supplies instead of a single one does not result in any change in the voltage gain expression.

• Input Resistance: The input resistance is given by $R_{in} = \dfrac{v_{in}(t)}{i_{in}(t)} \approx h_{11} + R_E\beta_F$ = 152 kΩ.

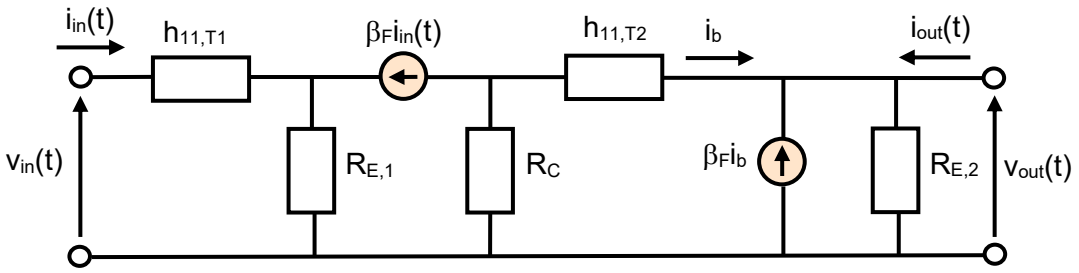
This is a very high input resistance (a feature that is highly desirable for a voltage amplifier in order to minimise the loading effect for the up-stream circuit). This input resistance is much larger than those of the CE amplifiers with negative feedback previously studied. This higher input resistance is due to the absence of both biasing resistances R_1 and R_2.

• Output Resistance: The output resistance is given by $R_{out} = \left.\dfrac{v_{out}(t)}{i_{out}(t)}\right|_{v_{in}(t)=0} = R_C$ = 6.2 kΩ (usual result for a CE amplifier).

Question 7: CE Amplifier with an Output Stage

We are not required to perform the DC analysis of the circuit. We can simply assume that the BJT is in the forward active mode without having to demonstrate it.

As for the AC analysis, the small-signal model of our CE-CC amplifier is depicted below.



The general equations for this circuit are as follows:

(1) $v_{in}(t) = \left[h_{11,T1} + (\beta_F + 1)R_{E,1}\right] i_{in}(t) \approx (h_{11,T1} + \beta_F R_{E,1}) i_{in}(t)$,

(2) $v_{out}(t) = -h_{11,T2}\, i_b - R_C\,(\beta_F i_{in}(t) + i_b) = -(h_{11,T2} + R_C) i_b - R_C \beta_F i_{in}(t)$,

and (3) $v_{out}(t) = R_{E,2}\,(i_{out}(t) + (\beta_F + 1) i_b) \approx R_{E,2}\,(i_{out}(t) + \beta_F\, i_b)$.

• OC Voltage Gain: The OC voltage gain of this amplifier is given by $A_v = \left.\dfrac{v_{out}(t)}{v_{in}(t)}\right|_{i_{out}(t)=0}$.

With $i_{out}(t) = 0$, the general equations for this circuit become

(1) $v_{in}(t) \approx (h_{11,T1} + \beta_F R_{E,1}) i_{in}(t)$,

(2) $v_{out}(t) = -(h_{11,T2} + R_C) i_b - R_C \beta_F i_{in}(t)$,

and (3) $v_{out}(t) \approx R_{E,2} \beta_F\, i_b$.

By combining these equations, we obtain the expression of the OC voltage gain of the CE-CC amplifier:

$$A_v \approx -\frac{R_{E,2} R_C (\beta_F)^2}{(R_C + h_{11,T2} + R_{E,2}\beta_F)(h_{11,T1} + R_{E,1}\beta_F)} \approx -3.1.$$

It is possible to somewhat simplify this expression by assuming that $R_C + h_{11,T2} << R_{E,2}\,\beta_F$ (i.e., 6.7 kΩ << 100 kΩ) and $h_{11,T1} << R_{E,1}\,\beta_F$ (i.e., 2 kΩ << 140 kΩ). In this case, we can write

$$A_v \approx -\frac{R_C}{R_{E,1}} \approx -3.4.$$

This expression is identical to that obtained with a simple CE amplifier with negative feedback. Therefore, we conclude that the use of a CC amplifier at the output of a CE amplifier does not result in any significant change in the OC voltage gain.

• Input Resistance: The input resistance is given by $R_{in} = \frac{v_{in}(t)}{i_{in}(t)} \approx h_{11,T1} + R_{E,1}\beta_F$ = 142 kΩ.

This is a very high input resistance. The expression of the input resistance is identical to that obtained for a classical CE amplifier with negative feedback and symmetrical power supplies. This implies that the presence of the CC amplifier does not have any effect on the input resistance of the overall circuit.

• Output Resistance: The output resistance is given by $R_{out} = \left.\frac{v_{out}(t)}{i_{out}(t)}\right|_{v_{in}(t)=0}$.

With $v_{in}(t)$ = 0, the general equations for this circuit become (1) $v_{out}(t) = -(h_{11,T2} + R_C) i_b$ and (2) $v_{out}(t) \approx R_{E,2}\,(i_{out}(t) + \beta_F\, i_b)$.

By combining both equations, we obtain the expression of the output resistance:

$$R_{out} \approx \frac{R_{E,2}(R_C + h_{11,T2})}{R_C + h_{11,T2} + R_{E,2}\,\beta_F} \approx 63\ \Omega.$$

It is possible to somewhat simplify this expression by assuming that $R_C + h_{11,T2} << R_{E,2}\,\beta_F$ (i.e., 6.7 kΩ << 100 kΩ). In this case, we can write

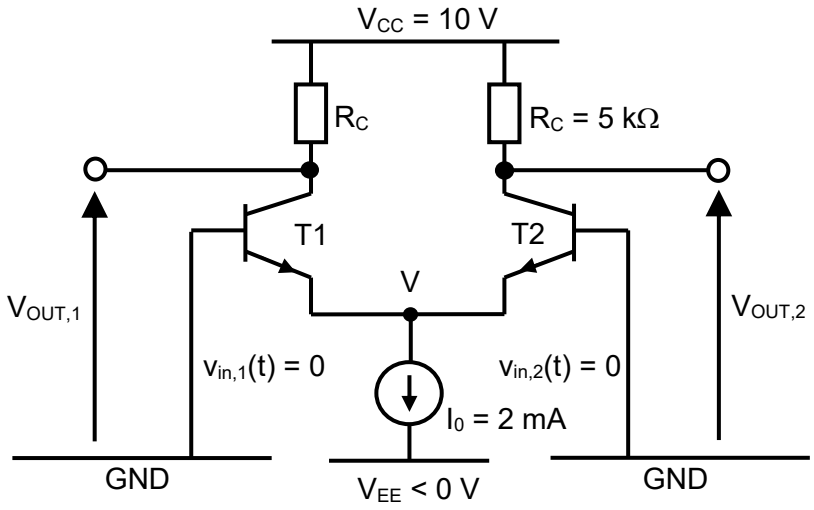
$$R_{out} \approx \frac{R_C + h_{11,T2}}{\beta_F} \approx 67\ \Omega.$$

Those results indicate that the output resistance is low (thanks to the presence of the CC amplifier). This is not the case with classical CE amplifiers for which $R_{out} = R_C$.

Therefore, due to their low output resistance (which is a very attractive feature for voltage amplifiers), CC amplifiers are often used at the output stage of more complex circuits, especially when the load resistance is low, in order to prevent an eventual collapse in the voltage gain (loading effect).

Question 8: Differential Amplifier

For the DC analysis, we consider the circuit below.



Assume that both transistors operate in the forward active mode (to be checked later).

The DC base-emitter voltages V_{BE1} and V_{BE2} of T1 and T2 are equal. Therefore, their base, collector, and emitter currents are also equal because T1 and T2 are assumed to be strictly identical. We thus have

$$I_{E1} = I_{E2} = \frac{I_0}{2} = 1 \text{ mA}.$$

Since T1 and T2 are assumed to be in the forward active mode of operation. In this case, we have

$$I_{C1} \approx I_{E1} = \frac{I_0}{2} \text{ and } I_{C2} \approx I_{E2} = \frac{I_0}{2},$$

thus yielding

$$V_{OUT,1} = V_{CC} - R_C I_{C1} \approx V_{CC} - R_C \frac{I_0}{2} = 5 \text{ volts}$$

and

$$V_{OUT,2} = V_{CC} - R_C I_{C2} \approx V_{CC} - R_C \frac{I_0}{2} = 5 \text{ volts}.$$

As $I_{B1} = I_{B2} \approx \frac{I_0}{2\beta_F} \approx 10\ \mu A$, the resistance h_{11} for both transistors is given by

$$h_{11} \approx \frac{2\beta_F V_T}{I_0} \approx 2.5\ k\Omega \text{ at room temperature.}$$

The emitter voltage V with respect to ground is equal to $0 - V_{BE,on} \approx -0.7$ volt.

Finally, we can tell whether or not the BJTs are in the forward active mode by computing

$$V_{CE1} = V_{OUT,1} - V \approx 5.7 \text{ volts}$$

and

$$V_{CE2} = V_{OUT,2} - V \approx 5.7 \text{ volts}.$$

We conclude that both transistors are clearly in the forward active mode.

The start of the cut-off mode corresponds to $I_{c1}(t) = I_{C1} + i_{c1}(t) = 0$ and $I_{c2}(t) = I_{C2} + i_{c2}(t) = 0$. In this case, we have $V_{out,1}(t) = V_{OUT,1} + v_{out,1}(t) = V_{CC}$ and $V_{out,2}(t) = V_{OUT,2} + v_{out,2}(t) = V_{CC}$. The DC operating point for both output voltages is therefore 5 volts away from the start of the cut-off mode.

The maximum collector current for each BJT is $I_0 = 2$ mA. This particular current value corresponds to $V_{out,1}(t) = V_{OUT,1} + v_{out,1}(t) = 0$ and $V_{out,2}(t) = V_{OUT,2} + v_{out,2}(t) = 0$. The DC operating point for both output voltages is once again 5 volts away from this extreme scenario.

Our conclusion is that both BJTs are ideally biased, thus ensuring a maximum output voltage swing equal to V_{CC} = 10 volts for both $V_{out,1}(t)$ and $V_{out,2}(t)$, and a voltage swing equal to $2V_{CC}$ = 20 volts for $v_{out}(t) = V_{out,1}(t) - V_{out,2}(t)$.

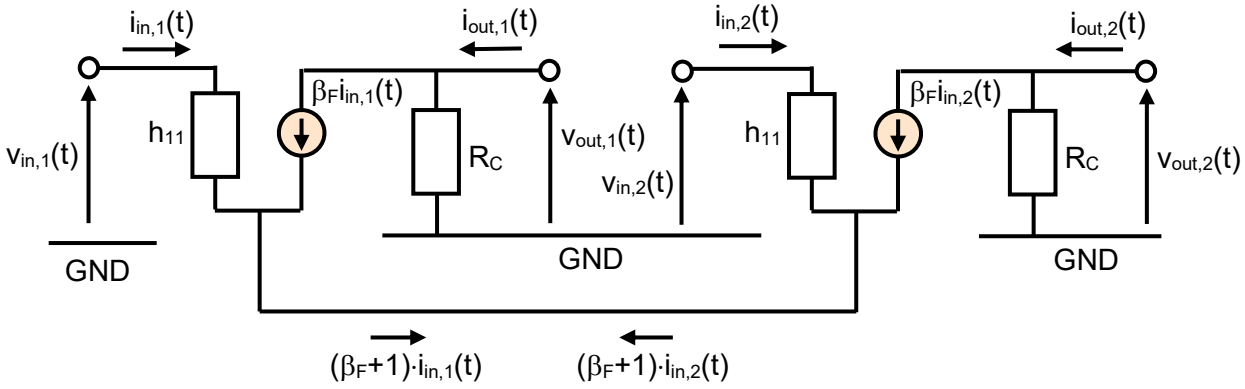
Before moving on to the AC analysis, we can ask ourselves whether T1 and T2 can ever enter the saturation mode.

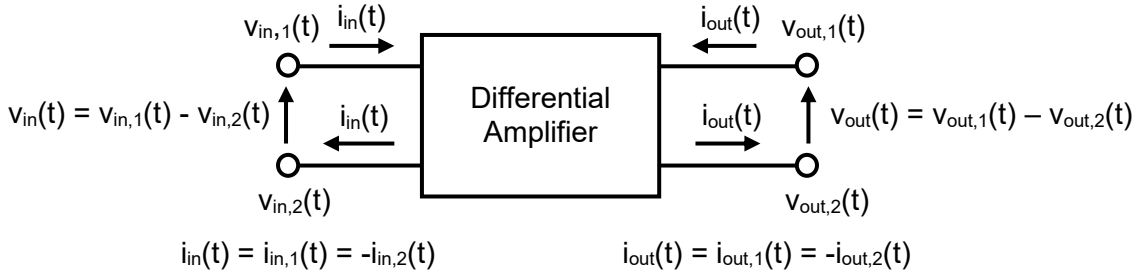
Consider, for instance, the BJT T1. This transistor T1 avoids the saturation mode as long as $V_{ce1}(t) = V_{out,1}(t) - (v_{in,1}(t) - V_{BE,on}) = V_{out,1}(t) + V_{BE,on} - v_{in,1}(t) > V_{CE,sat}$, i.e. $v_{in,1}(t) < V_{out,1}(t) + V_{BE,on} - V_{CE,sat}$.

Considering the worst-case scenario, the minimum possible value for $V_{out,1}(t)$ has just been shown to be equal to 0 volt. In this case, T1 avoids the saturation mode as long as $v_{in,1}(t) < V_{BE,on} - V_{CE,sat} \approx 500$ mV.

We conclude that we must ensure that, at all times, both conditions $v_{in,1}(t) < 500$ mV and $v_{in,2}(t) < 500$ mV are met in order to prevent T1 and/or T2 from ever going into saturation.

As for the AC analysis, the small-signal model of our differential amplifier is depicted below.





The general equations for this circuit are as follows:

(1) $v_{in,1}(t) - h_{11} i_{in,1}(t) = v_{in,2}(t) - h_{11} i_{in,2}(t)$, which is equivalent to $v_{in,1}(t) - v_{in,2}(t) = 2h_{11} i_{in}(t)$ since $i_{in}(t) = i_{in,1}(t) = -i_{in,2}(t)$,

(2) $v_{out,1}(t) = R_C \left(i_{out,1}(t) - \beta_F i_{in,1}(t)\right)$, i.e. $v_{out,1}(t) = R_C \left(i_{out}(t) - \beta_F i_{in}(t)\right)$,

and (3) $v_{out,2}(t) = R_C \left(i_{out,2}(t) - \beta_F i_{in,2}(t)\right)$, i.e. $v_{out,2}(t) = -R_C \left(i_{out}(t) - \beta_F i_{in}(t)\right)$.

• OC Voltage Gain: The OC voltage gain of the differential amplifier is given by

$$A_v = \left.\frac{v_{out}(t)}{v_{in}(t)}\right|_{i_{out}(t)=0} = \left.\frac{v_{out,1}(t) - v_{out,2}(t)}{v_{in,1}(t) - v_{in,2}(t)}\right|_{i_{out}(t)=0}.$$

With $i_{out}(t) = 0$, the general equations for this circuit become

(1) $v_{in,1}(t) - v_{in,2}(t) = 2h_{11} i_{in}(t)$, (2) $v_{out,1}(t) = -R_C \beta_F i_{in}(t)$, and (3) $v_{out,2}(t) = R_C \beta_F i_{in}(t)$.

By combining these equations, we obtain the expression of the OC voltage gain of the differential amplifier:

$$A_v = -\frac{R_C \beta_F}{h_{11}} \approx \text{-200, at room temperature.}$$

Note that the OC voltage gain can also be expressed as

$$A_v = -\frac{R_C \beta_F}{h_{11}} = -\frac{R_C \beta_F I_0}{2\beta_F V_T} = -\frac{R_C I_0}{2V_T}.$$

• Input Resistance: The input resistance is given by

$$R_{in} = \frac{v_{in}(t)}{i_{in}(t)} = \frac{v_{in,1}(t) - v_{in,2}(t)}{i_{in}(t)} = 2h_{11} \approx 5\ \text{k}\Omega \text{ at room temperature.}$$

• Output Resistance: The output resistance is given by

$$R_{out} = \left.\frac{v_{out}(t)}{i_{out}(t)}\right|_{v_{in}(t)=0} = \left.\frac{v_{out,1}(t) - v_{out,2}(t)}{i_{out}(t)}\right|_{v_{in,1}(t) = v_{in,2}(t)}.$$

With $v_{in,1}(t) = v_{in,2}(t)$, the general equations for this circuit become

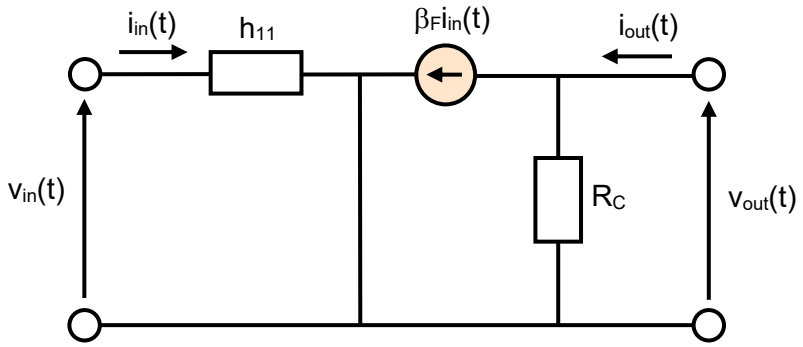
(1) $v_{in,1}(t) - v_{in,2}(t) = 2h_{11}i_{in}(t) = 0$, which is equivalent to $i_{in}(t) = 0$,

(2) $v_{out,1}(t) = R_C\, i_{out}(t)$, and (3) $v_{out,2}(t) = -R_C\, i_{out}(t)$.

We finally obtain $R_{out} = 2R_C$ = 10 kΩ.

Question 9: Characteristics of Generic BJT Amplifiers

(a) Common-emitter (CE) amplifier



The general equations for this circuit are as follows:

$$(1)\ v_{in}(t) = h_{11}i_{in}(t) \text{ and } (2)\ v_{out}(t) = R_C\left(i_{out}(t) - \beta_F i_{in}(t)\right).$$

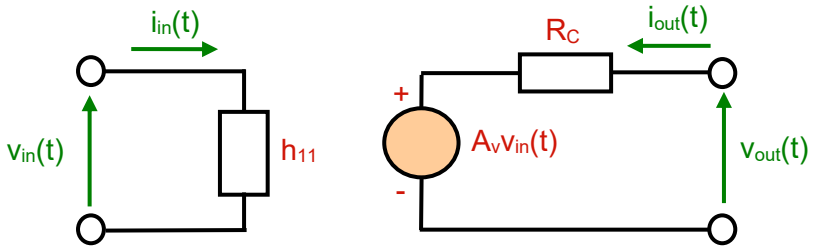
• OC reverse voltage gain: $A_{v,in} = \left.\dfrac{v_{in}(t)}{v_{out}(t)}\right|_{i_{in}(t)=0} = 0 \rightarrow$ The CE amplifier has a unilateral voltage amplifier model. ☺

• OC voltage gain: $A_v = \left.\dfrac{v_{out}(t)}{v_{in}(t)}\right|_{i_{out}(t)=0} = -\beta_F \dfrac{R_C}{h_{11}} \rightarrow$ High OC voltage gain. ☺

• Input resistance: $R_{in} = \dfrac{v_{in}(t)}{i_{in}(t)} = h_{11} \rightarrow$ Medium input resistance.

• Output resistance: $R_{out} = \left.\dfrac{v_{out}(t)}{i_{out}(t)}\right|_{v_{in}(t)=0} = R_C \rightarrow$ Medium output resistance.

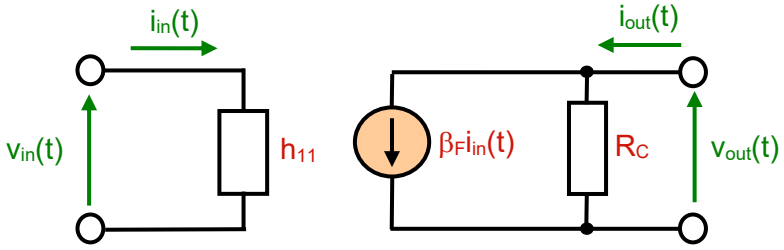
The voltage amplifier model of the CE amplifier is shown below.



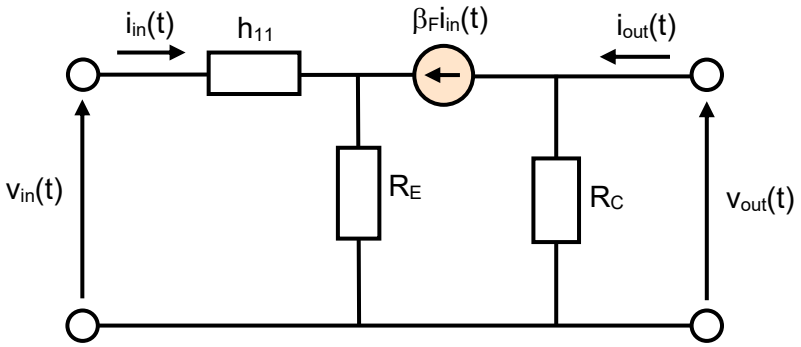
As for the current amplifier model of the CE amplifier, we know that it is unilateral with $R_{in} = h_{11}$ and $R_{out} = R_C$. Hence, what remains to be determined is only the SC current gain A_i given by

$$A_i = \left.\frac{i_{out}(t)}{i_{in}(t)}\right|_{v_{out}(t)=0} = \beta_F \rightarrow \text{High SC current gain. } \ddot{\smile}$$

The current amplifier model of the CE amplifier is shown below.



(b) CE amplifier with negative feedback



The general equations for this circuit are as follows:

$$(1)\ v_{in}(t) = [h_{11} + (\beta_F + 1)R_E]i_{in}(t) \text{ and } (2)\ v_{out}(t) = R_C(i_{out}(t) - \beta_F i_{in}(t)).$$

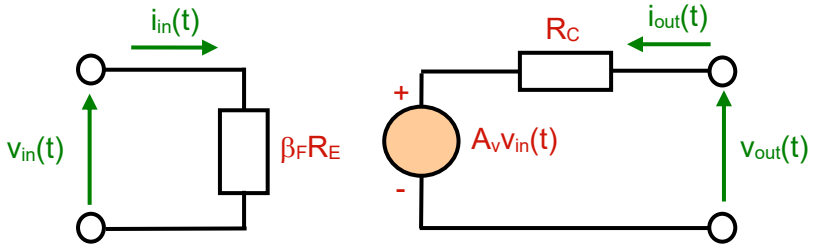
- OC reverse voltage gain: $A_{v,in} = \left.\frac{v_{in}(t)}{v_{out}(t)}\right|_{i_{in}(t)=0} = 0 \rightarrow$ The CE amplifier with negative feedback has a unilateral voltage amplifier model. ☺

- OC voltage gain: $A_v = \left.\frac{v_{out}(t)}{v_{in}(t)}\right|_{i_{out}(t)=0} = -\frac{\beta_F R_C}{h_{11} + (\beta_F + 1)R_E} \approx -\frac{R_C}{R_E} \rightarrow$ Low-to-medium OC voltage gain that does not depend on β_F.

- Input resistance: $R_{in} = \frac{v_{in}(t)}{i_{in}(t)} = h_{11} + (\beta_F + 1)R_E \approx \beta_F R_E \rightarrow$ Large input resistance. ☺

- Output resistance: $R_{out} = \left.\frac{v_{out}(t)}{i_{out}(t)}\right|_{v_{in}(t)=0} = R_C \rightarrow$ Medium output resistance.

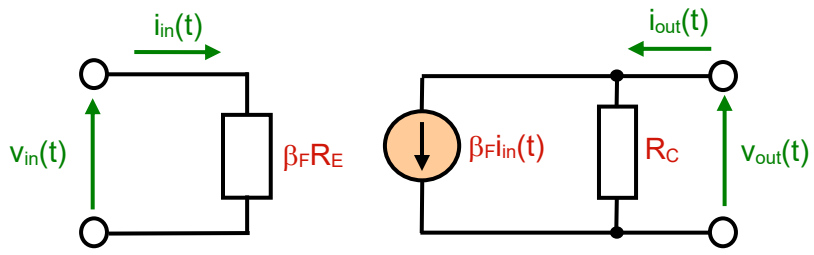
The voltage amplifier model of the CE amplifier with negative feedback is shown below.



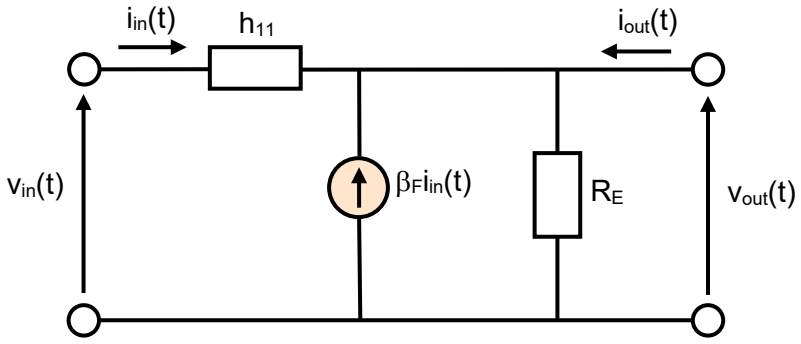
As for the current amplifier model of the CE amplifier with negative feedback, we know that it is unilateral with $R_{in} \approx \beta_F R_E$ (large input resistance → good for a voltage amplifier, bad for a current amplifier) and $R_{out} = R_C$. Hence, what remains to be determined is only the SC current gain A_i given by

$$A_i = \left.\frac{i_{out}(t)}{i_{in}(t)}\right|_{v_{out}(t)=0} = \beta_F \rightarrow \text{High SC current gain. ☺}$$

The current amplifier model of the CE amplifier with negative feedback is shown below.



(c) Common-collector (CC) amplifier



The general equations for this circuit are as follows:

$$(1)\ v_{in}(t) = h_{11}\, i_{in}(t) + v_{out}(t) \text{ and } (2)\ v_{out}(t) = R_E\left(i_{out}(t) + (\beta_F + 1) i_{in}(t)\right).$$

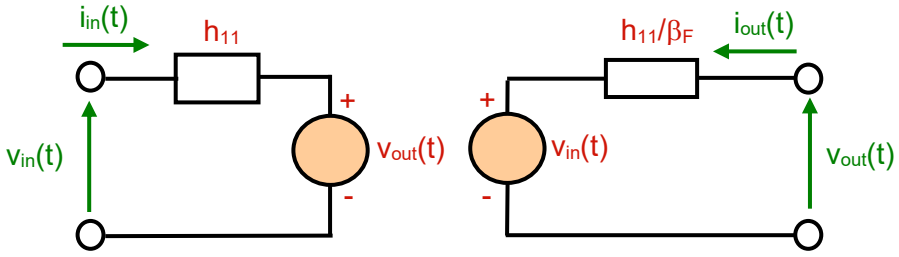
• OC reverse voltage gain: $A_{v,in} = \left.\dfrac{v_{in}(t)}{v_{out}(t)}\right|_{i_{in}(t)=0} = 1 \rightarrow$ The CC amplifier does not have a unilateral voltage amplifier model. We must therefore determine its bilateral voltage amplifier model instead. ☹

• OC forward voltage gain: $A_{v,out} = \left.\dfrac{v_{out}(t)}{v_{in}(t)}\right|_{i_{out}(t)=0} = -\dfrac{(\beta_F + 1) R_E}{h_{11} + (\beta_F + 1) R_E} \approx 1 \rightarrow$ Unit OC forward voltage gain. ☹

• Input resistance: $R_{in} = \left.\frac{v_{in}(t)}{i_{in}(t)}\right|_{v_{out}(t)=0} = h_{11}$ → Medium input resistance.

• Output resistance: $R_{out} = \left.\frac{v_{out}(t)}{i_{out}(t)}\right|_{v_{in}(t)=0} = \frac{h_{11}R_E}{h_{11}+(\beta_F+1)R_E} \approx \frac{h_{11}}{\beta_F}$ → Low output resistance. ☺

The voltage amplifier model of the CC amplifier is shown below.



As for the current amplifier model of the CC amplifier, there is no shortcut in this case because it has to be bilateral, exactly like the voltage amplifier model. We thus need to restart the whole procedure to find this current amplifier model.

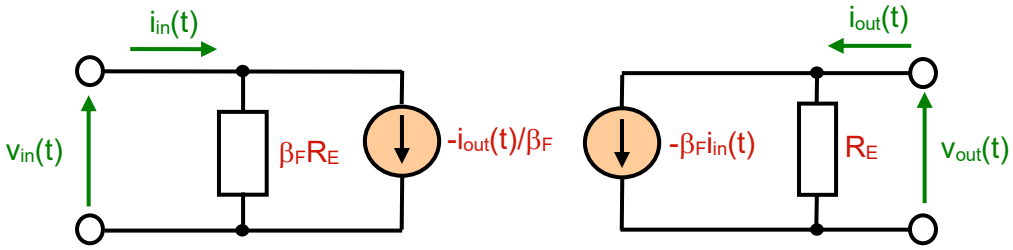
• SC reverse current gain: $A_{i,in} = \left.\frac{i_{in}(t)}{i_{out}(t)}\right|_{v_{in}(t)=0} = -\frac{R_E}{h_{11}+(\beta_F+1)R_E} \approx -\frac{1}{\beta_F}$ → The CC amplifier does indeed have a bilateral current amplifier model. ☹

• SC forward current gain: $A_{i,out} = \left.\frac{i_{out}(t)}{i_{in}(t)}\right|_{v_{out}(t)=0} = -(\beta_F+1) \approx -\beta_F$ → High SC forward current gain. ☺

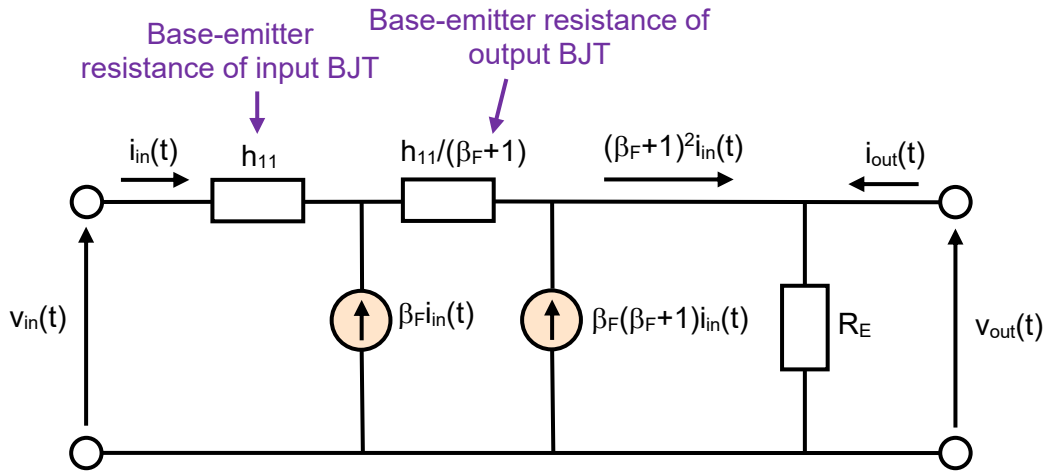
• Input resistance: $R_{in} = \left.\frac{v_{in}(t)}{i_{in}(t)}\right|_{i_{out}(t)=0} = h_{11}+(\beta_F+1)R_E \approx \beta_F R_E$ → High input resistance. ☹

• Output resistance: $R_{out} = \left.\frac{v_{out}(t)}{i_{out}(t)}\right|_{i_{in}(t)=0} = R_E$ → Medium output resistance.

The current amplifier model of the CC amplifier is shown below.



(d) Darlington-pair amplifier (CC-CC configuration)



The general equations for this circuit are as follows:

(1) $v_{in}(t) = h_{11} i_{in}(t) + \dfrac{h_{11}(\beta_F + 1)}{\beta_F + 1} i_{in}(t) + v_{out}(t) = 2h_{11} i_{in}(t) + v_{out}(t)$,

and (2) $v_{out}(t) = R_E\left(i_{out}(t) + (\beta_F + 1)^2 i_{in}(t)\right)$.

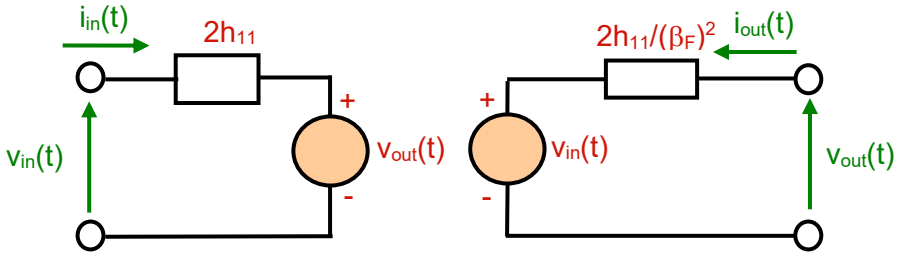
• OC reverse voltage gain: $A_{v,in} = \left.\dfrac{v_{in}(t)}{v_{out}(t)}\right|_{i_{in}(t)=0} = 1 \rightarrow$ The Darlington-pair amplifier does not have a unilateral voltage amplifier model. We must therefore determine its bilateral voltage amplifier model instead. ☹

• OC forward voltage gain: $A_{v,out} = \left.\frac{v_{out}(t)}{v_{in}(t)}\right|_{i_{out}(t)=0} = -\frac{(\beta_F+1)^2 R_E}{2h_{11}+(\beta_F+1)^2 R_E} \approx 1$ → Unit OC forward voltage gain. ☹

• Input resistance: $R_{in} = \left.\frac{v_{in}(t)}{i_{in}(t)}\right|_{v_{out}(t)=0} = 2h_{11}$ → Medium input resistance.

• Output resistance: $R_{out} = \left.\frac{v_{out}(t)}{i_{out}(t)}\right|_{v_{in}(t)=0} = \frac{2h_{11}R_E}{2h_{11}+(\beta_F+1)^2 R_E} \approx \frac{2h_{11}}{(\beta_F)^2}$ → Very low output resistance. ☺

The voltage amplifier model of the Darlington-pair amplifier is shown below. Note that this model is very similar to that of a CC amplifier, which is not surprising as the Darlington-pair amplifier can be viewed as an improved version of a traditional CC amplifier.



As for the current amplifier model of the Darlington-pair amplifier, there is no shortcut in this case because it has to be bilateral, exactly like the voltage amplifier model. We thus need to restart the whole procedure to find this current amplifier model.

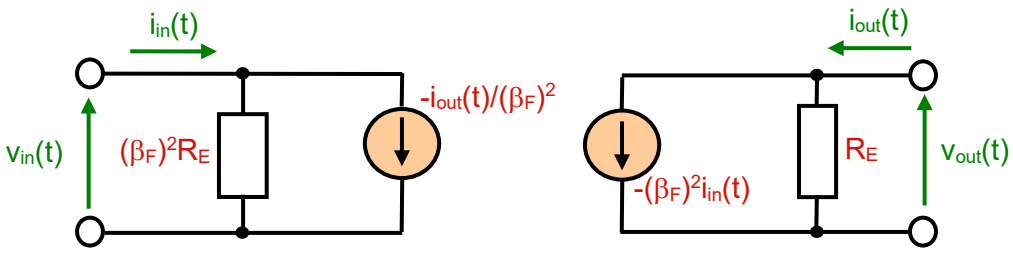
• SC reverse current gain: $A_{i,in} = \left.\frac{i_{in}(t)}{i_{out}(t)}\right|_{v_{in}(t)=0} = -\frac{R_E}{2h_{11}+(\beta_F+1)^2 R_E} \approx -\frac{1}{(\beta_F)^2}$ → The Darlington-pair amplifier does indeed have a bilateral current amplifier model. ☹

• SC forward current gain: $A_{i,out} = \left.\frac{i_{out}(t)}{i_{in}(t)}\right|_{v_{out}(t)=0} = -(\beta_F+1)^2 \approx -(\beta_F)^2$ → High SC forward current gain. ☺

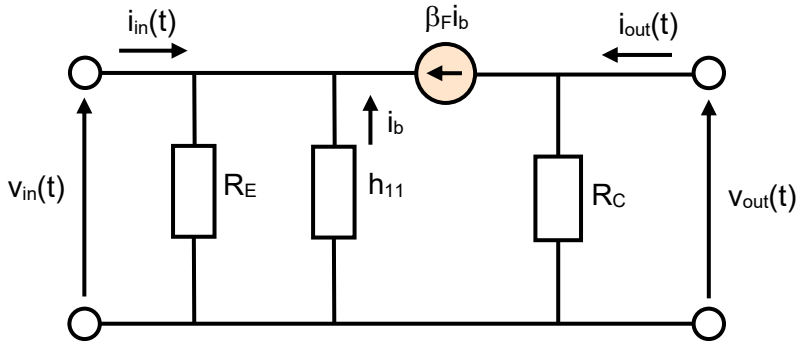
• Input resistance: $R_{in} = \left.\frac{v_{in}(t)}{i_{in}(t)}\right|_{i_{out}(t)=0} = 2h_{11} + (\beta_F + 1)^2 R_E \approx (\beta_F)^2 R_E \rightarrow$ Very high input resistance. ☹

• Output resistance: $R_{out} = \left.\frac{v_{out}(t)}{i_{out}(t)}\right|_{i_{in}(t)=0} = R_E \rightarrow$ Medium output resistance.

The current amplifier model of the Darlington-pair amplifier is shown below.



(e) Common-base (CB) amplifier



The general equations for this circuit are as follows:

(1) $v_{in}(t) = -h_{11} i_b$, (2) $v_{in}(t) = R_E[i_{in}(t) + (\beta_F + 1) i_b]$, and (3) $v_{out}(t) = R_C(i_{out}(t) - \beta_F i_b)$.

• OC reverse voltage gain: $A_{v,in} = \left.\frac{v_{in}(t)}{v_{out}(t)}\right|_{i_{in}(t)=0} = 0$.

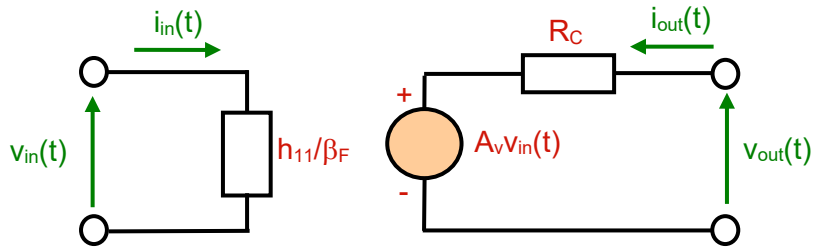
→ The CB amplifier has a unilateral voltage amplifier model. ☺

• OC voltage gain: $A_v = \left.\frac{v_{out}(t)}{v_{in}(t)}\right|_{i_{out}(t)=0} = \frac{\beta_F R_C}{h_{11}}$ → High OC voltage gain (identical to that obtained with the CE amplifier, but with a positive sign). ☺

• Input resistance: $R_{in} = \frac{v_{in}(t)}{i_{in}(t)} = \frac{h_{11} R_E}{h_{11} + (\beta_F + 1) R_E} \approx \frac{h_{11}}{\beta_F}$ → Low input resistance. ☹

• Output resistance: $R_{out} = \left.\frac{v_{out}(t)}{i_{out}(t)}\right|_{v_{in}(t)=0} = R_C$ → Medium output resistance.

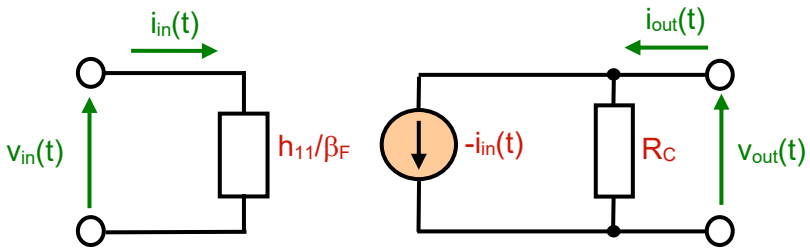
The voltage amplifier model of the CB amplifier is shown below.



As for the current amplifier model of the CB amplifier, we know that it is unilateral with $R_{in} = h_{11}/\beta_F$ (low input resistance → bad for a voltage amplifier, good for a current amplifier) and $R_{out} = R_C$. Hence, what remains to be determined is only the SC current gain A_i given by

$$A_i = \left.\frac{i_{out}(t)}{i_{in}(t)}\right|_{v_{out}(t)=0} = -\frac{\beta_F R_E}{h_{11} + (\beta_F + 1) R_E} \approx -1$$ → Unit SC current gain. ☹

The current amplifier model of the CB amplifier is shown below.



Type of amplifier	Voltage amplifier model	$A_{v,in}$	$A_{v,out}$	A_v	R_{in}	R_{out}
CE	Unilateral	0	-	$-\beta_F \frac{R_C}{h_{11}}$	h_{11}	R_C
CE with Feedback	Unilateral	0	-	$-\frac{R_C}{R_E}$	$\beta_F R_E$	R_C
CC	Bilateral	1	1	-	h_{11}	$\frac{h_{11}}{\beta_F}$
Darlington (CC-CC)	Bilateral	1	1	-	$2h_{11}$	$\frac{2h_{11}}{(\beta_F)^2}$
CB	Unilateral	0	-	$\beta_F \frac{R_C}{h_{11}}$	$\frac{h_{11}}{\beta_F}$	R_C

Type of amplifier	Current amplifier model	$A_{i,in}$	$A_{i,out}$	A_i	R_{in}	R_{out}
CE	Unilateral	0	-	β_F	h_{11}	R_C
CE with Feedback	Unilateral	0	-	β_F	$\beta_F R_E$	R_C
CC	Bilateral	$-\frac{1}{\beta_F}$	$-\beta_F$	-	$\beta_F R_E$	R_E
Darlington (CC-CC)	Bilateral	$-\frac{1}{(\beta_F)^2}$	$-(\beta_F)^2$	-	$(\beta_F)^2 R_E$	R_E
CB	Unilateral	0	-	-1	$\frac{h_{11}}{\beta_F}$	R_C

- END -